

Lecture 7: Numeric Prediction

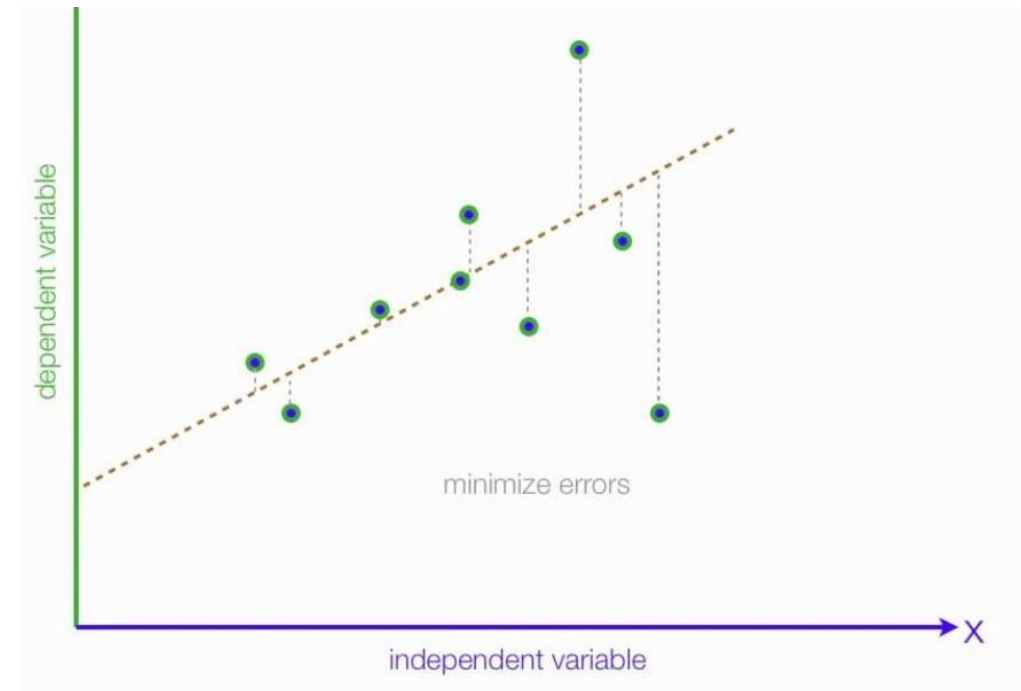
Outline

- Numeric Prediction
- Regression Methods
 - Simple Linear Regression
 - Multiple Linear Regression
 - Regression Trees

Numeric Prediction

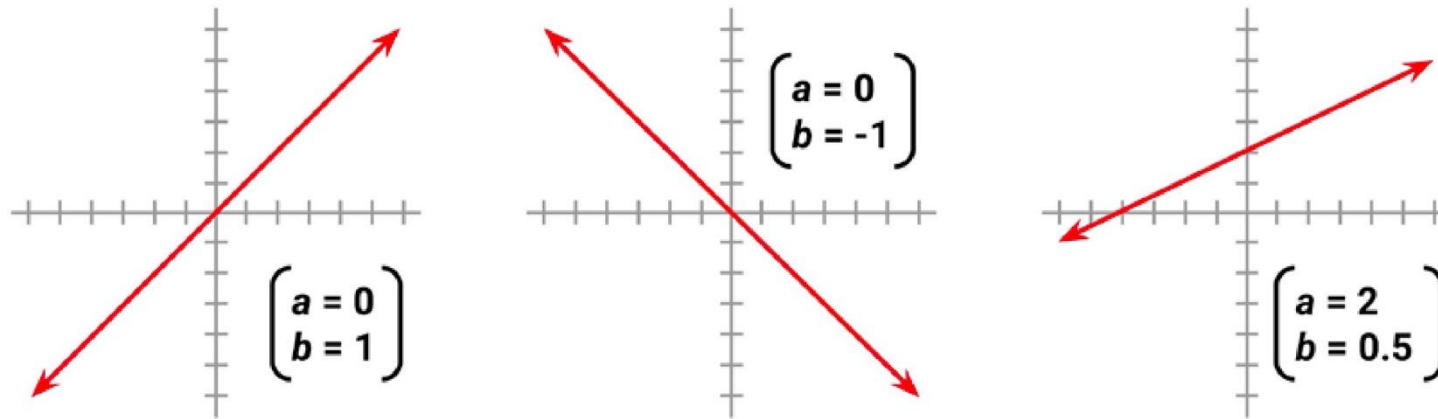
The construction and evaluation of models used to generate **predictions of numeric values of a target variable.**

- **Supervised learning-Numerical Prediction**
 - If a company's sales have increased steadily every month for the past few years, by conducting a linear analysis on the sales data with monthly sales, the company could forecast sales in future months.



Regression Methods

- Regression is concerned with specifying relationships between a **single numeric dependent variable** (the target variable) and **one or more independent variables** (the predictors)



- Lines can be defined in a slope-intercept form: $y = a + bx$
 - The letter y indicates the **dependent variable** and x indicates the **independent variable**
 - The **slope term** b specifies how much the line rises for each increase in x
 - Positive values defines lines that slope upward while negative values define lines that slope downward.
 - The term a is known as the **intercept** because it specifies the point where the line crosses, or intercepts the vertical y axis.

Simple Linear Regression

Predicting Medical Expenses – Example

- Predicting medical expenses using linear regression
 - In order for a health insurance company to make money, it needs to collect more in yearly premium than it spends on medical care to its beneficiaries.
 - Insurance companies develop models that can accurately forecast medical expenses for its clients.
 - Goal: use patient data to estimate the medical care expenses given the health condition of a patient.
 - The estimate can be used to set the price of yearly premiums.

Simple Linear Regression

- Insurance data

	age	sex	bmi	children	smoker	region	expenses
0	19	female	27.9	0	yes	southwest	16884.92
1	18	male	33.8	1	no	southeast	1725.55
2	28	male	33.0	3	no	southeast	4449.46
3	33	male	22.7	0	no	northwest	21984.47
4	32	male	28.9	0	no	northwest	3866.86
...	...	...	...	...	...	...	...
1333	50	male	31.0	3	no	northwest	10600.55
1334	18	female	31.9	0	no	northeast	2205.98
1335	18	female	36.9	0	no	southeast	1629.83
1336	21	female	25.8	0	no	southwest	2007.95
1337	61	female	29.1	0	yes	northwest	29141.36

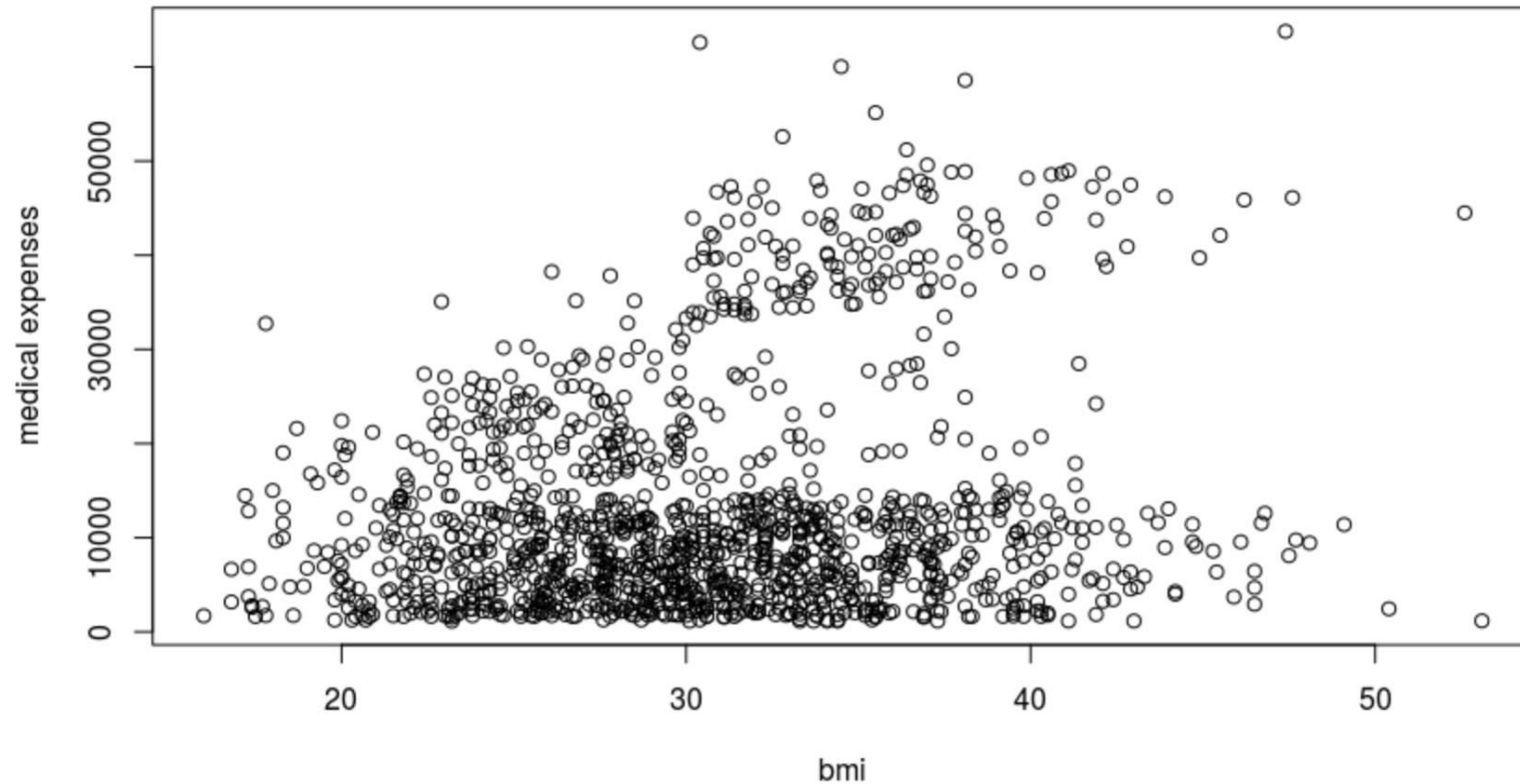
1338 rows x 7 columns

Simple Linear Regression

- Simple Linear Regression
 - Dependent variable: expenses
 - Independent variable: bmi

Simple Linear Regression

- Scatterplot of bmi and expenses



Simple Linear Regression

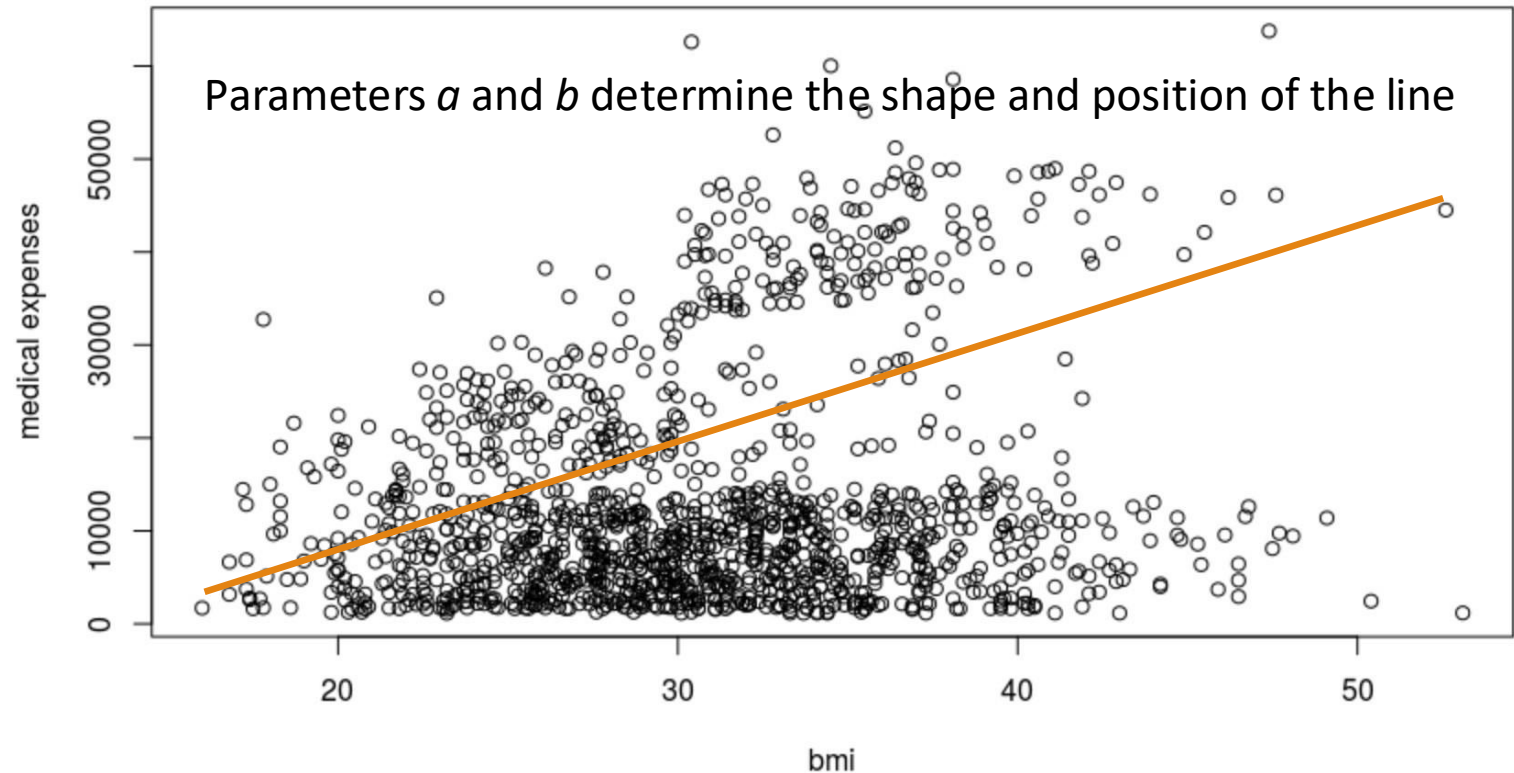
- To model the relationship between bmi and expenses, we can turn to simple linear regression.
- A simple linear regression model defines the relationship between a dependent variable and a single independent variable using a line defined by an equation in the following form:

$$y = a + bx$$

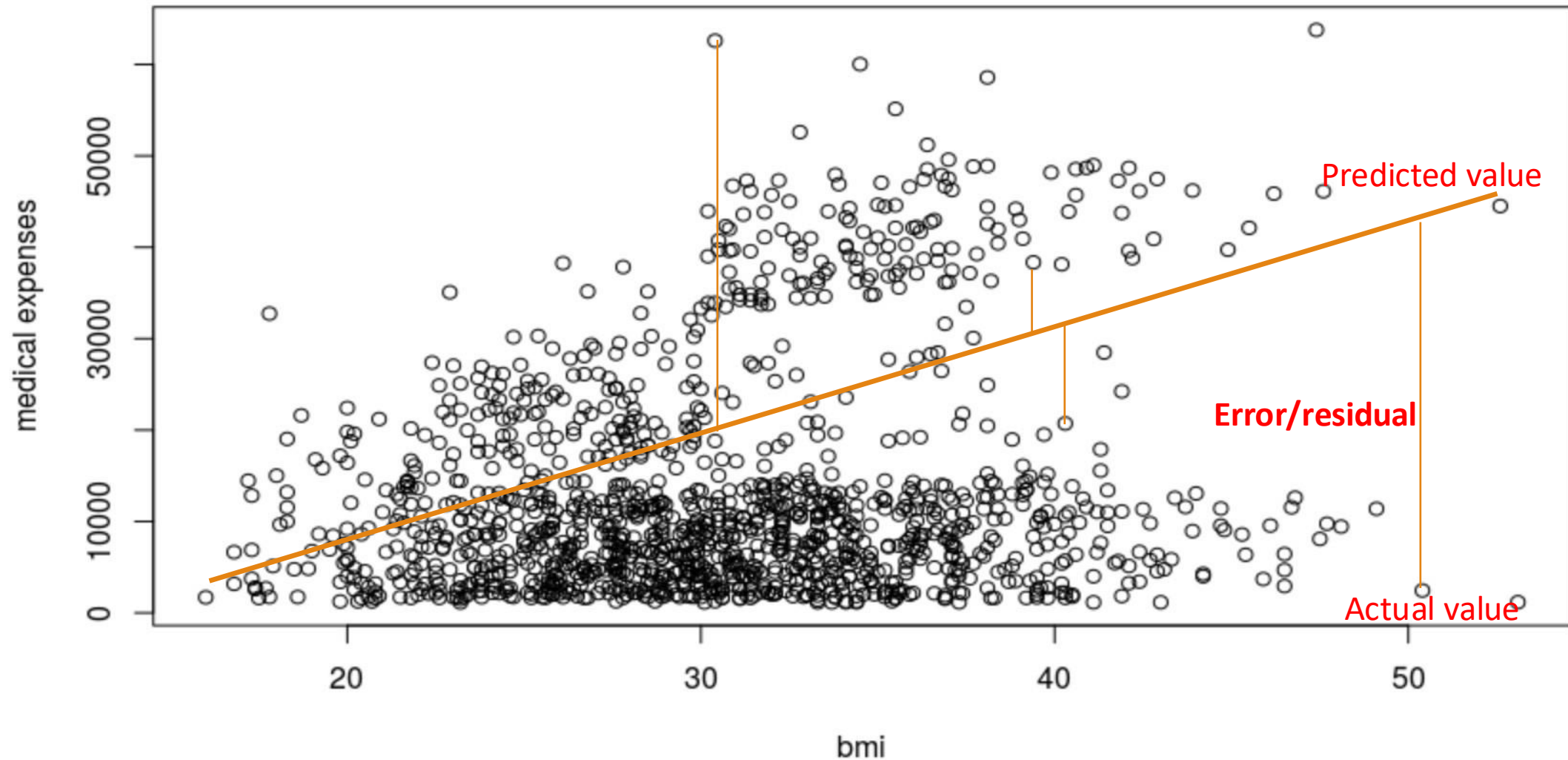
Simple Linear Regression

- Assume: $\text{expenses} = 785.25 + 409.90 * \text{bmi}$

The line doesn't pass through each data point exactly. Instead, it cuts through the data somewhat evenly, with some predictions lower or higher than the line.



Ordinary Least Squares (OLS) Estimation



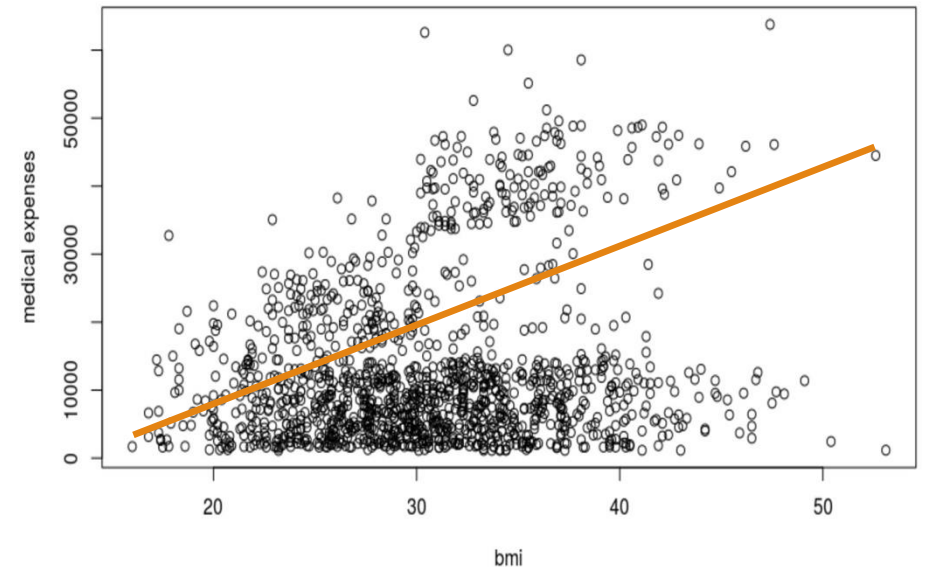
Ordinary Least Squares (OLS) Estimation

- In order to determine the optimal estimates of a and b , an estimation method known as **Ordinary Least Squares (OLS)** was used.

- Minimize the residuals: $\sum (y_i - \hat{y}_i)^2 = \sum e_i^2$

$$a = \bar{y} - b\bar{x} \quad b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

age	sex	bmi	children	smoker	region	expenses
19	female	27.9	0	yes	southwest	16884.92
18	male	33.8	1	no	southeast	1725.55
28	male	33	3	no	southeast	4449.46
33	male	22.7	0	no	northwest	21984.47
32	male	28.9	0	no	northwest	3866.86
31	female	25.7	0	no	southeast	3756.62
46	female	33.4	1	no	southeast	8240.59
37	female	27.7	3	no	northwest	7281.51
37	male	29.8	2	no	northeast	6406.41
60	female	25.8	0	no	northwest	28923.14
25	male	26.2	0	no	northeast	2721.32



Multiple Linear Regression

Multiple Linear Regression

- Multiple linear regression is an extension of simple linear regression.
 - $y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \beta_ix_i + \varepsilon$
 - It has additional terms for multiple independent variables
 - The dependent variable y is specified as the sum of an intercept term β_0 plus the product of the estimated β value and the x values for each of the i features.
- The goal: find values of **beta coefficients** that minimize the prediction error of a linear equation.

Multiple Linear Regression

- Step 1: Prepare data for prediction

	age	sex	bmi	children	smoker	region	expenses
0	19	female	27.9	0	yes	southwest	16884.92
1	18	male	33.8	1	no	southeast	1725.55
2	28	male	33.0	3	no	southeast	4449.46
3	33	male	22.7	0	no	northwest	21984.47
4	32	male	28.9	0	no	northwest	3866.86
...	...	...	...	...	...	...	...
1333	50	male	31.0	3	no	northwest	10600.55
1334	18	female	31.9	0	no	northeast	2205.98
1335	18	female	36.9	0	no	southeast	1629.83
1336	21	female	25.8	0	no	southwest	2007.95
1337	61	female	29.1	0	yes	northwest	29141.36

1338 rows x 7 columns

Dummy Coding

- Convert categorical variables into numeric format – **dummy coding**
- Create dummy variables to replace the original categorical variable.
- Dummy coding for a binary variable: gender

$$\text{male} = \begin{cases} 1 & \text{if } x = \text{male} \\ 0 & \text{otherwise} \end{cases}$$

Color: Red, Pink, Blue		<i>n</i> categories, <i>n</i> -1 dummy variables		
	Color		Color_Red	Color_Pink
Diamond1	Red		1	0
Diamond2	Pink		0	1
Diamond3	Blue		0	0

Multiple Linear Regression

- Step2: splitting data to training and testing sets

	age	bmi	children	expenses	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.9	0	16884.92	0	1	0	0	1
1	18	33.8	1	1725.55	1	0	0	1	0
2	28	33.0	3	4449.46	1	0	0	1	0
3	33	22.7	0	21984.47	1	0	1	0	0
4	32	28.9	0	3866.86	1	0	1	0	0
...	...	...	...	...	...	...	...	...	...
931	39	32.5	1	6238.30	0	0	0	0	1
932	46	25.8	5	10096.97	1	0	0	0	1
933	45	35.3	0	7348.14	0	0	0	0	1
934	32	37.2	2	4673.39	1	0	0	1	0
...	...	...	...	...	...	...	...	...	...
1333	50	31.0	3	10600.55	1	0	1	0	0
1334	18	31.9	0	2205.98	0	0	0	0	0
1335	18	36.9	0	1629.83	0	0	0	1	0
1336	21	25.8	0	2007.95	0	0	0	0	1
1337	61	29.1	0	29141.36	0	1	1	0	0

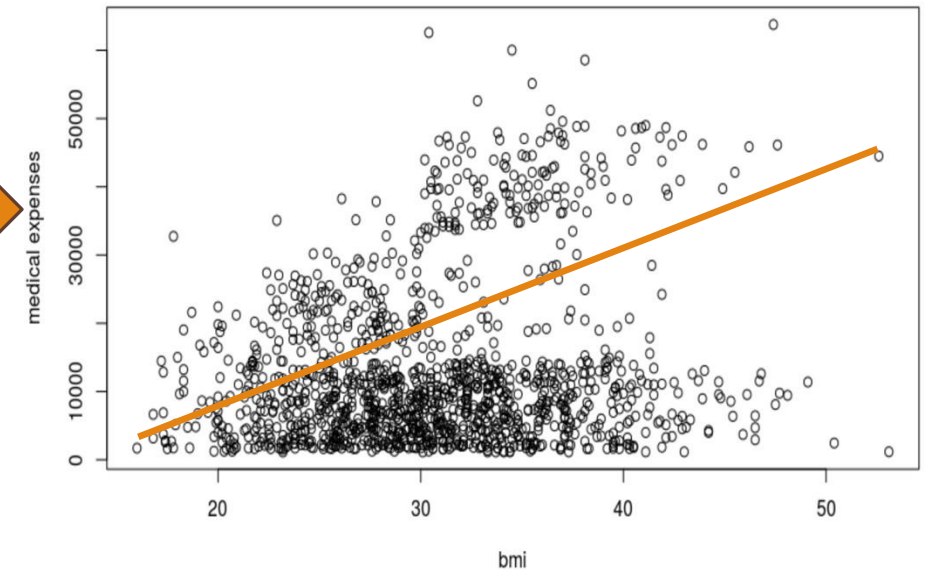
70% training data

30% testing data

Multiple Linear Regression

- Step3: Train Multiple Linear Regression model on **training data (70%)**

	age	bmi	children	expenses	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.9	0	16884.92	0	1	0	0	1
1	18	33.8	1	1725.55	1	0	0	1	0
2	28	33.0	3	4449.46	1	0	0	1	0
3	33	22.7	0	21984.47	1	0	1	0	0
4	32	28.9	0	3866.86	1	0	1	0	0
...	...	...	...	...	...	...	...	...	...
931	39	32.5	1	6238.30	0	0	0	0	1
932	46	25.8	5	10096.97	1	0	0	0	1
933	45	35.3	0	7348.14	0	0	0	0	1
934	32	37.2	2	4673.39	1	0	0	1	0
935	59	27.5	0	12233.83	0	0	0	0	1



$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_i x_i + \varepsilon$$

Calculate the coefficients

Multiple Linear Regression

- 1. The **residuals** section provides summary statistics for the errors in our predictions.
 - **Residual = real value – predicted value**
 - 50 percent of errors fall within the 1Q and 3Q values (the first and third quartile), so the majority of predictions were between \$2,988.446 over the true value and \$1,343.2315 under the true value.

1

Residuals:

Min	1Q	Median	3Q	Max
-30302.4407	-1343.2315	985.6509	2988.446	11148.5211

2

Coefficients:

	Estimate	Std. Error	t value	p value
_intercept	-11836.117831	1191.289921	-9.9355	0.000000
age	256.402924	10.186903	25.1699	0.000000
bmi	335.414456	16.372545	20.4864	0.000000
children	472.939931	166.843442	2.8346	0.004687
sex_male	-47.692891	400.980436	-0.1189	0.905348
smoker_yes	23435.046128	501.984750	46.6848	0.000000
region_northwest	-561.353117	545.419326	-1.0292	0.303645
region_southeast	-995.537125	551.901974	-1.8038	0.071580
region_southwest	-798.824371	541.710520	-1.4746	0.140648

3

R-squared: 0.73096, Adjusted R-squared: 0.72864
F-statistic: 314.82 on 8 features

Multiple Linear Regression

- 2. The beta coefficients indicate the estimated increase in expenses for an increase of one in each of the variables, assuming all other values are held constant.
 - Intercept is of little value alone because it is impossible to have values of zero for all features

Expenses = -11836.117831 + 256.402924*age + 335.414456*bmi + 472.939931*children - 47.692891*sexmale + 23435.046128*smokeryes - 561.353117*regionnorthwest - 995.537125* regionsoutheast - 798.824371* regionsouthwest

Residuals: 1				
Min	1Q	Median	3Q	Max
-30302.4407	-1343.2315	985.6509	2988.446	11148.5211
Coefficients: 2				
	Estimate	Std. Error	t value	p value
_intercept	-11836.117831	1191.289921	-9.9355	0.000000
age	256.402924	10.186903	25.1699	0.000000
bmi	335.414456	16.372545	20.4864	0.000000
children	472.939931	166.843442	2.8346	0.004687
sex_male	-47.692891	400.980436	-0.1189	0.905348
smoker_yes	23435.046128	501.984750	46.6848	0.000000
region_northwest	-561.353117	545.419326	-1.0292	0.303645
region_southeast	-995.537125	551.901974	-1.8038	0.071580
region_southwest	-798.824371	541.710520	-1.4746	0.140648

R-squared: 0.73096, Adjusted R-squared: 0.72864 3				
F-statistic: 314.82 on 8 features				

Multiple Linear Regression

- 2. For each estimated regression coefficient, the **p-value**, provides an estimate of the probability that the true coefficient is zero given the value of the estimate.
 - Small p-values suggest that the true coefficient is very unlikely to be zero, which means that the variable is extremely unlikely to have no relationship with the dependent variable.
 - p-values less than the significance level (**<0.05**) are considered **statistically significant**.

1

Residuals:

Min	1Q	Median	3Q	Max
-30302.4407	-1343.2315	985.6509	2988.446	11148.5211

2

Coefficients:

	Estimate	Std. Error	t value	p value
_intercept	-11836.117831	1191.289921	-9.9355	0.000000
age	256.402924	10.186903	25.1699	0.000000
bmi	335.414456	16.372545	20.4864	0.000000
children	472.939931	166.843442	2.8346	0.004687
sex_male	-47.692891	400.980436	-0.1189	0.905348
smoker_yes	23435.046128	501.984750	46.6848	0.000000
region_northwest	-561.353117	545.419326	-1.0292	0.303645
region_southeast	-995.537125	551.901974	-1.8038	0.071580
region_southwest	-798.824371	541.710520	-1.4746	0.140648

3

R-squared: 0.73096, Adjusted R-squared: 0.72864

F-statistic: 314.82 on 8 features

Multiple Linear Regression

- 3. The **multiple R-squared value** provides a measure of how well our model as a whole explains the values of the dependent variable.
 - The model explains nearly 73 percent of the variation in the dependent variable.
 - **High R-squared on training data indicates overfitting.**

1

Residuals:

Min	1Q	Median	3Q	Max
-30302.4407	-1343.2315	985.6509	2988.446	11148.5211

2

Coefficients:

	Estimate	Std. Error	t value	p value
_intercept	-11836.117831	1191.289921	-9.9355	0.000000
age	256.402924	10.186903	25.1699	0.000000
bmi	335.414456	16.372545	20.4864	0.000000
children	472.939931	166.843442	2.8346	0.004687
sex_male	-47.692891	400.980436	-0.1189	0.905348
smoker_yes	23435.046128	501.984750	46.6848	0.000000
region_northwest	-561.353117	545.419326	-1.0292	0.303645
region_southeast	-995.537125	551.901974	-1.8038	0.071580
region_southwest	-798.824371	541.710520	-1.4746	0.140648

3

R-squared: 0.73096, Adjusted R-squared: 0.72864
F-statistic: 314.82 on 8 features

Multiple Linear Regression

• 4. Other Metrics

Linear Regression Assumptions:

1. Linearity: The relationship is linear.

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \varepsilon$$

2. Independence: Observations (errors) are independent. $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0 (i \neq j)$

3. Homoscedasticity: Constant variance of errors. $\text{Var}(\varepsilon|X) = \sigma^2$

4. No perfect multicollinearity:

Predictors are not perfectly correlated.

5. Zero conditional mean

(exogeneity): Errors are uncorrelated with predictors. $E(\varepsilon|X) = 0$

6. Normality of errors: Residuals are normally distributed (mainly for inference). $\varepsilon \sim N(0, \sigma^2)$

OLS Regression Results

Dep. Variable:	expenses	R-squared:	0.731
Model:	OLS	Adj. R-squared:	0.729
Method:	Least Squares	F-statistic:	314.8
Date:	Wed, 11 Mar 2026	Prob (F-statistic):	3.47e-258
Time:	01:51:48	Log-Likelihood:	-9495.3
No. Observations:	936	AIC:	1.901e+04
Df Residuals:	927	BIC:	1.905e+04
Df Model:	8		
Covariance Type:	nonrobust		

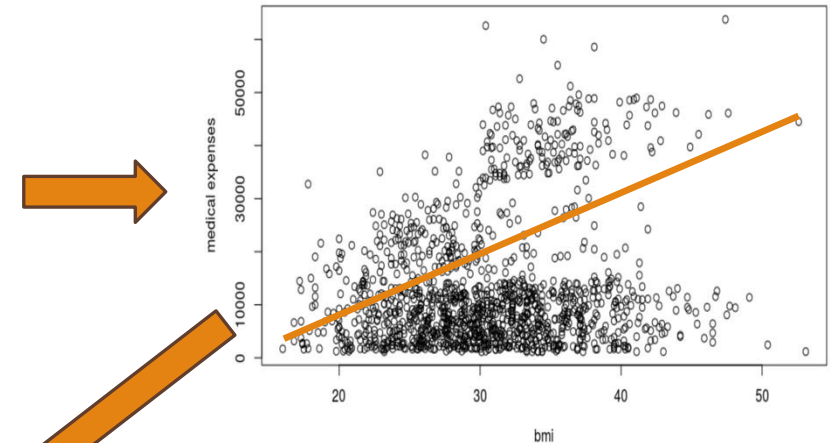
	coef	std err	t	P> t	[0.025	0.975]
const	-1.184e+04	1205.932	-9.814	0.000	-1.42e+04	-9468.636
age	256.4354	14.628	17.531	0.000	227.728	285.143
bmi	335.3691	34.437	9.738	0.000	267.785	402.954
children	472.7098	168.734	2.802	0.005	141.565	803.854
sex_male	-47.5468	405.909	-0.117	0.907	-844.154	749.060
smoker_yes	2.344e+04	507.660	46.165	0.000	2.24e+04	2.44e+04
region_northwest	-561.9011	590.574	-0.951	0.342	-1720.918	597.116
region_southeast	-994.7487	591.668	-1.681	0.093	-2155.912	166.414
region_southwest	-799.4489	585.816	-1.365	0.173	-1949.129	350.231

Omnibus:	232.849	Durbin-Watson:	2.048	2.
Prob(Omnibus):	0.000	Jarque-Bera (JB):	571.734	
Skew:	1.309	Prob(JB):	7.07e-125	
Kurtosis:	5.794	Cond. No.	315.	4.

Multiple Linear Regression

- Step 4: make predictions on **testing data (30%)** data (70%)

	age	bmi	children	expenses	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.9	0	16884.92	0	1	0	0	1
1	18	33.8	1	1725.55	1	0	0	1	0
2	28	33.0	3	4449.46	1	0	0	1	0
3	33	22.7	0	21984.47	1	0	1	0	0
4	32	28.9	0	3866.86	1	0	1	0	0
...	...	...	...	...	...	...	...	...	...
931	39	32.5	1	6238.30	0	0	0	0	1
932	46	25.8	5	10096.97	1	0	0	0	1
933	45	35.3	0	7348.14	0	0	0	0	1
934	32	37.2	2	4673.39	1	0	0	1	0
...	...	...	...	...	...	...	...	...	...
1333	50	31.0	3		1	0	1	0	0
1334	18	31.9	0		0	0	0	0	0
1335	18	36.9	0		0	0	0	1	0
1336	21	25.8	0		0	0	0	0	1
1337	61	29.1	0		0	1	1	0	0



Multiple Linear Regression

- Step 4: make predictions on **testing data (30%)** data (70%)

expenses	Predicted expenses
16884.92	15445.54
1725.55	1455.21
4449.46	6799.98
21984.47	12351.13
3866.86	2434.12
3756.62	1235.12
8240.59	6394.12
7281.51	5450.74
6406.41	2509.23
28923.14	39455.26
2721.32	2311.65



Examine prediction performance

Evaluation Metrics for Numeric Prediction

- **Mean Absolute Error (MAE)**

- The **mean absolute error (MAE)** is a quantity used to measure how close forecasts or predictions are to the eventual outcomes.
- This is known as a scale-dependent accuracy measure and therefore cannot be used to make comparisons between series using different scales

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|$$

- **Root Mean Squared Error (RMSE)**

- **RMSE** gives a relatively high weight to large errors. This means the RMSE should be more useful when large errors are particularly undesirable.

$$\text{RMSE} = \sqrt{\frac{\sum_{t=1}^n (y_t - \hat{y}_t)^2}{n}}$$

MAE: 4011.3808850910846
RMSE: 5773.7111375700415

Improving Model Performance: Adding non-linear relationships

- In linear regression, the relationship between independent variable and the dependent variable is assumed to be **linear**. Yet this may not necessarily be true.
 - The effect of age on medical cost may not be **constant** throughout all the age values
 - To account for a non-linear relationship, we can add a higher order term to the regression model. We will be modeling relationship like this:

$$y = \alpha + \beta_1 x + \beta_2 x^2$$

Improving Model Performance: Adding non-linear relationships

Residuals:

	Min	1Q	Median	3Q	Max
	-30897.6196	-1143.0599	976.6449	3010.6995	10996.9905

Coefficients:

	Estimate	Std. Error	t value	p value
_intercept	-7946.613879	2024.274416	-3.9257	0.000093
age	27.144543	49.795286	0.5451	0.585799
bmi	333.788744	21.173262	15.7646	0.000000
children	591.183643	171.920444	3.4387	0.000610
sex_male	-70.422281	400.892935	-0.1757	0.860596
smoker_yes	23462.056874	502.406607	46.6993	0.000000
region_northwest	-526.692320	548.948589	-0.9595	0.337577
region_southeast	-967.924415	550.533067	-1.7582	0.079048
region_southwest	-781.301189	542.270220	-1.4408	0.149977
age2	2.888628	0.108131	26.7142	0.000000

R-squared: 0.73257, Adjusted R-squared: 0.72997

F-statistic: 281.85 on 9 features

Expenses = -7946.613879 + 27.144543*age - 70.422281*sexmale + 333.788744*bmi + 591.183643*children + 23462.056874*smokeryes - 526.692320*regionnorthwest – 967.924415* regionsoutheast – 781.301189* regionsouthwest + 2.888628 * age2

Increase age from 20 to 30

Increase age from 40 to 50

Improving Model Performance: Adding non-linear relationships

Residuals:

	Min	1Q	Median	3Q	Max
	-30897.6196	-1143.0599	976.6449	3010.6995	10996.9905

Coefficients:

	Estimate	Std. Error	t value	p value
_intercept	-7946.613879	2024.274416	-3.9257	0.000093
age	27.144543	49.795286	0.5451	0.585799
bmi	333.788744	21.173262	15.7646	0.000000
children	591.183643	171.920444	3.4387	0.000610
sex_male	-70.422281	400.892935	-0.1757	0.860596
smoker_yes	23462.056874	502.406607	46.6993	0.000000
region_northwest	-526.692320	548.948589	-0.9595	0.337577
region_southeast	-967.924415	550.533067	-1.7582	0.079048
region_southwest	-781.301189	542.270220	-1.4408	0.149977
age2	2.888628	0.108131	26.7142	0.000000

R-squared: 0.73257, Adjusted R-squared: 0.72997

F-statistic: 281.85 on 9 features

Expenses = -7946.613879 + 27.144543*age - 70.422281*sexmale + 333.788744*bmi + 591.183643*children + 23462.056874*smokeryes - 526.692320*regionnorthwest - 967.924415*regionsoutheast - 781.301189* regionsouthwest + 2.888628 * age2

Increase age from 20 to 30

- Expenses will be increased by
(900 - 400) * **2.888628** + (30-20)* **27.144543**

Increase age from 40 to 50

- Expenses will be increased by
(2500 - 1600) * **2.888628** + (50-40)* **27.144543**

Improving Model Performance:

Converting a numeric variable into a binary indicator

- The effect of a numeric feature may not be cumulative, rather it has an effect only after a specific threshold has been reached.
 - BMI may have zero impact on medical expenditures for individuals in the normal weight range
 - It may be strongly related to higher costs for the obese (that is, BMI of 30 or above).
- Create a binary obesity indicator variable that is 1 if the BMI is at least 30, and 0 if less.
 - The estimated beta for this binary feature would then indicate the average net impact on medical expenses for individuals with BMI of 30 or above, relative to those with BMI less than 30

Improving Model Performance:

Converting a numeric variable into a binary indicator

Residuals:

Min	1Q	Median	3Q	Max
-29380.2963	-1285.7016	-36.9711	3447.0453	11900.2978

Coefficients:

	Estimate	Std. Error	t value	p value
_intercept	-4233.850435	2176.564771	-1.9452	0.052051
age	50.008498	50.645108	0.9874	0.323687
bmi	153.166557	22.771932	6.7261	0.000000
children	596.007862	170.344625	3.4988	0.000489
sex_male	-71.755309	397.162085	-0.1807	0.856666
smoker_yes	23458.777167	497.650302	47.1391	0.000000
region_northwest	-535.310285	543.848095	-0.9843	0.325222
region_southeast	-874.057659	545.540885	-1.6022	0.109452
region_southwest	-824.907342	537.106870	-1.5358	0.124917
age2	2.576038	0.107299	24.0079	0.000000
bmi30	2810.337867	625.035974	4.4963	0.000008

R-squared: 0.73766, Adjusted R-squared: 0.73482

F-statistic: 260.10 on 10 features

Improving Model Performance: Adding Interaction Effects

- What if certain features have a combined impact on the dependent variable?
 - For instance, smoking and obesity may have harmful effects separately.
 - Their combined effect may be worse than the sum of each one alone.
- When two features have a combined effect, this is known as an **interaction**.

Improving Model Performance: Adding Interaction Effects

```
Residuals:
    Min       1Q   Median       3Q      Max
-24356.6801  818.38  1311.4334  1694.2262  4055.3453

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1334.525875   1638.900791  -0.8143  0.415691
age           45.644502     38.078226   1.1987  0.230947
bmi          124.927286     17.127101   7.2941  0.000000
children     602.951131     127.997651   4.7106  0.000003
sex_male    -323.859152     298.512453  -1.0849  0.278241
smoker_yes  13444.538555     471.284949  28.5274  0.000000
region_northwest -565.712615    408.660324  -1.3843  0.166594
region_southeast -1022.186642    409.954486  -2.4934  0.012824
region_southwest -1158.326763    403.636849  -2.8697  0.004201
age2           2.808353       0.080626  34.8318  0.000000
bmi30        -1000.934145    483.373489  -2.0707  0.038658
bmi30_smoker  20016.357007    701.429309  28.5365  0.000000
---
R-squared:  0.85189,    Adjusted R-squared:  0.85012
F-statistic: 483.13 on 11 features
```

**Expenses = - 1334.525875 – 45.644502*age - 323.859152
*sexmale + 124.927286 *bmi + 602.951131 *children +
13444.538555 *smokeryes – 565.712615*regionnorthwest –
1022.186642* regionsoutheast – 1158.326763*
regionsouthwest + 2.808353 * age2 - 1000.934145 *bmi30 +
20016.357007 *(smokeryes *bmi30)**

For non-obese patients:

- Smoker increases the expenses by **13444.538555**

For obese patients:

- Smoker increases the expenses by **13444.538555 + 20016.357007**

Improving Model Performance: Adding Interaction Effects

Add a higher-order "age" term:

MAE: 3953.7602463146113
RMSE: 5707.448582070192

Add a binary indicator:

MAE: 4086.8961912203076
RMSE: 5658.92764301156

Add an interaction effect:

MAE: 2309.1294296071137
RMSE: 4097.094074358147

An Improved Regression Model

- Based on domain knowledge of how medical costs may be related to patient characteristics, we developed what we think is a more accurately specified regression formula.
- To summarize the improvements, we:
 - Added a non-linear term for age
 - Created an indicator for obesity
 - Specified an interaction between obesity and smoking

Regression Trees

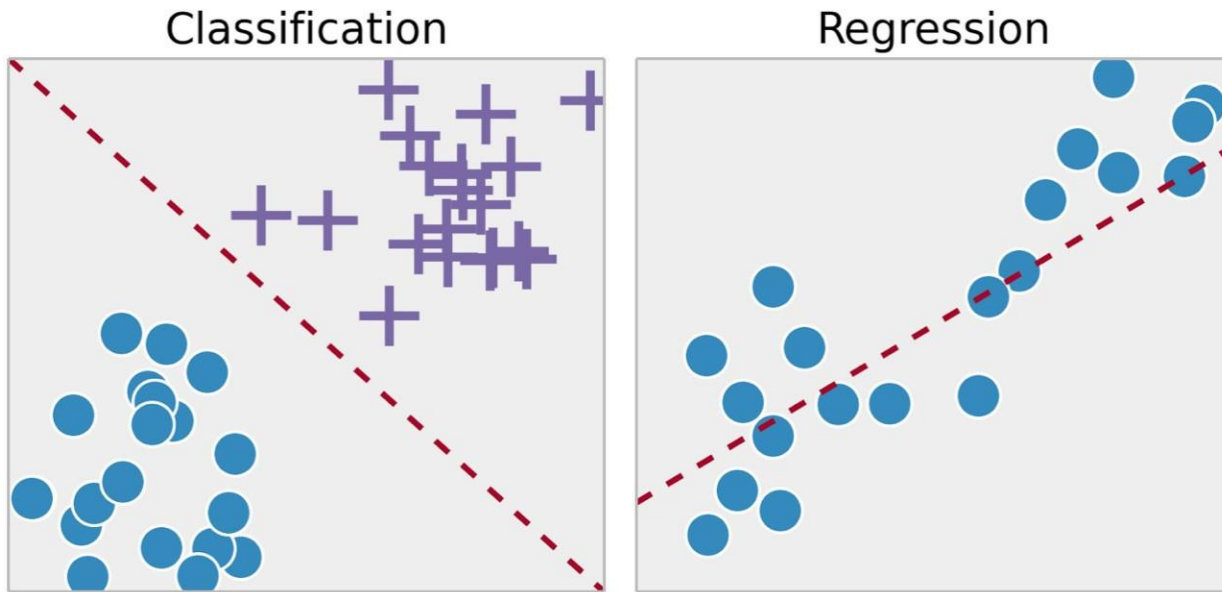
Trees for Numeric Prediction

- A decision tree can also be used for numeric prediction by making only small adjustments to the tree-growing algorithm.
- Trees for numeric prediction:
 - Regression trees

Numeric Prediction vs Classification

- **Numerical Prediction**

- A numerical prediction problem is when the output variable is a real value, such as “dollars” or “weight”.



Regression Tree

- Regression tree
 - Trees for numeric prediction are built in much the same way as they are for classification.
 - Predicted value of each leaf node is typically reported as mean value of the target variable for all training observations belonging to node.
 - SDR (Standard deviation reduction) or squared error reduction is used to choose a predictor to split a node, similar to the way classification error is used in decision tree. Decision trees may be better suited for tasks with many complex, non-linear relationships among predictors and target variable.

Regression Tree

Reduce the mean squared error most

